



(KTXFI2EBNF) Physics II. Lecture

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November 24, 2025

Common Business I

- ▶ I had time to translate the Hungarian text in the images into English; however, I have not yet had time to upload the updated figures to Moodle. I will do so soon.
- ▶ I have also not yet had time to replace the Hungarian text directly on the images with English labels. I apologize; I will update them shortly.
- ▶ We will not have sufficient time to cover the following topics:
 - ▶ Basic concepts of nuclear physics.
 - ▶ Basic concepts of particle physics.
- ▶ I can only briefly outline what these topics include: radioactivity, isotopes, neutron radiation, radiation protection, neutron-induced nuclear fission, thermonuclear fusion, nuclear batteries, types of nuclear reactors, thermonuclear experiments, nuclear weapons, thermonuclear weapons, management of nuclear waste, the Hungarian Nuclear Power Plant in Paks, and future technologies such as advanced reactors, high-temperature reactors, thorium reactors, and modular reactors.

Common Business II

- ▶ What could I meaningfully tell you about particle physics at this point? For example: hadrons, mesons, leptons, quarks, or particle colliders?
- ▶ We are out of time; therefore, I will send you additional links, or I may record and upload a short personal video exclusively for you.
- ▶ Naturally, none of these topics will be included in the midterm or final examinations.

Where are we? I

- ▶ ~~Motion of charged particles in electromagnetic fields.~~
- ▶ ~~Elements of quantum mechanics. Heisenberg's uncertainty principle. The stationary Schrödinger equation and its applications.~~
- ▶ ~~Limits of the classical conceptual framework. Thermal radiation. Photoelectric effect. Compton effect. The dual nature of electromagnetic radiation. The dual nature of particles.~~
- ▶ ~~Moving reference frames. Inertial forces in accelerating reference frames. Elements of special relativity. Dirac equation, antimatter.~~
- ▶ ~~The classical theory of atomic structure (Rutherford, Franck-Hertz experiment, Bohr model, quantum numbers, Pauli exclusion principle).~~
- ▶ ~~Physics of condensed matter. Metallic bonding. Electrical conduction in metals based on the free electron model and the wave model. Hall effect. Band theory of solids.~~

Where are we? II

- ▶ Semiconductors. Elements of Fermi-Dirac statistics. Thermoelectric phenomena. Magnetic properties.
- ▶ Ferroelectricity. Piezoelectricity and electrostriction. Liquid crystals. Superconductivity.
- ▶ **Luminescence. Lasers.** ~~Basic knowledge of nuclear physics. Basic knowledge of particle physics.~~

Luminescence, Fluorescence, Phosphorescence I

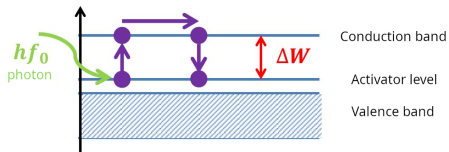
- ▶ Luminescence is the emission of visible light by certain materials when exposed to visible light, X-rays, or radioactive radiation.
- ▶ Unlike thermal radiation, this emission appears even at low temperatures.
- ▶ Luminescence: The emission of light in cases where the origin of the emitted radiation is not the temperature of the emitting body is called luminescence. Materials that emit such radiation - i.e., materials that luminesce - are referred to as phosphors.
- ▶ Mechanisms inducing luminescence:
 - ▶ Light - photoluminescence.
 - ▶ Electrical phenomena - electroluminescence.
 - ▶ Radioactive radiation - radioluminescence.
 - ▶ Biological processes - bioluminescence (fireflies, certain bacteria).
- ▶ Luminescence occurs when a material absorbs incoming radiation and subsequently re-emits it, typically at a different wavelength.

Luminescence, Fluorescence, Phosphorescence II

- ▶ Duration of luminescence: The time interval during which a material continues to emit light after the cessation of the excitation is called the luminescence lifetime.
- ▶ Luminescent vs. non-luminescent materials:
 - ▶ Pure crystals with few defects do **not** luminesce.
 - ▶ Crystal defects are required for luminescence to occur (impurities often enhance the effect).
 - ▶ Impurities that facilitate the process are called *activators*.
 - ▶ PHOSPHORS = Base material (ZnS, CaS, CdS, KCl, NaCl) + Activator (Ag, Cu, Bi, Mn, Tl)

Luminescence, Fluorescence, Phosphorescence III

- Fluorescence is the form of luminescence in which emission ceases almost immediately after the excitation stops (typically $< 10^{-6}$ s). In short: the light emission effectively ends as soon as the excitation is removed.



- An incoming photon elevates an electron of the activator atom from the activator level to the conduction band.
- This electron can move freely within the crystal.
- If, during its motion, it encounters another activator atom with an available electron site, it recombines — i.e., it returns to the activator level.
- The lifetime of the excited state determines the duration of emitted light.

Luminescence, Fluorescence, Phosphorescence IV

- ▶ Phosphorescence: Luminescence in which emission persists for a significantly longer duration — longer than 10^{-5} – 10^{-4} s — is called phosphorescence. In practice, emission may continue for several minutes after excitation ceases.
- ▶ A recombination-inhibiting mechanism must be present. For example, impurities or lattice defects may create “traps” within the crystal.
- ▶ These traps can capture electrons wandering through the crystal, preventing recombination with the activator atom.
- ▶ Traps correspond to energetically favorable states immediately below the conduction band.
- ▶ Electrons remain trapped until they receive sufficient energy from lattice vibrations (phonons) to return to the conduction band. They may then become trapped again... thus, an electron may undergo multiple trapping cycles.
- ▶ Eventually, upon encountering an activator atom, recombination occurs and luminescent light is emitted.

Luminescence, Fluorescence, Phosphorescence V

- ▶ Stokes' theorem: The frequency of fluorescence light cannot exceed — and its wavelength cannot be shorter than — the frequency or wavelength of the exciting radiation.
- ▶ For the incoming photon:

$$hf_0 \geq \Delta W.$$

For the emitted photon:

$$hf = \Delta W.$$

Thus:

$$hf_0 \geq hf \quad \Rightarrow \quad f_0 \geq f.$$

Using $f = \frac{c}{\lambda}$:

$$h \frac{c}{\lambda_0} \geq h \frac{c}{\lambda},$$

Luminescence, Fluorescence, Phosphorescence VI

which yields:

$$\frac{1}{\lambda_0} \geq \frac{1}{\lambda}.$$

Therefore:

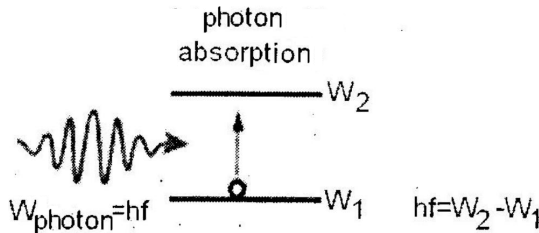
$$\lambda_0 \leq \lambda.$$

q. e. d.

- ▶ Applications of phosphorescence:
 - ▶ Wavelength conversion: e.g., interior coatings of mercury-vapor lamps or fluorescent tubes.
 - ▶ CRT television screens.
 - ▶ Oscilloscope screens.
 - ▶ Clock dials.
 - ▶ etc. . . .

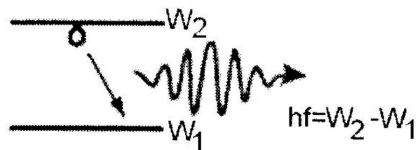
Absorption, Emission, Stimulated Emission, LASER I

Absorption occurs when atoms absorb energy and transition to higher energy states.



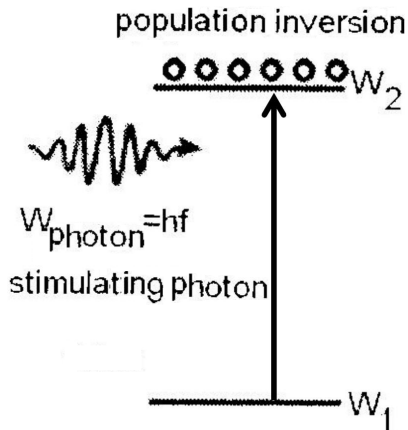
Spontaneous emission is the natural decay of excited electrons accompanied by photon emission.

photon emission
(spontaneous emission)



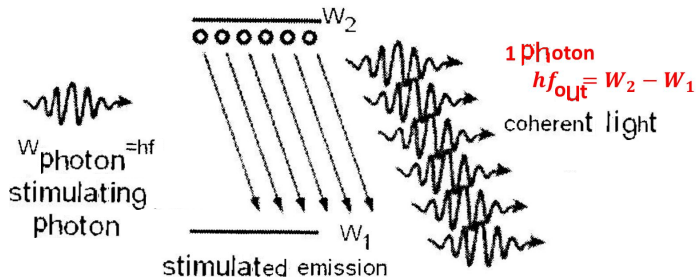
Absorption, Emission, Stimulated Emission, LASER II

Population Inversion: A phenomenon in which more electrons occupy the higher-energy level W_2 than the lower-energy level W_1 . In this case, the electron population of the higher energy state becomes “more populated” than that of the lower energy state. This situation is contrary to the principle of energy minimization. Hence the name: *population inversion*.



Absorption, Emission, Stimulated Emission, LASER III

Induced or stimulated emission



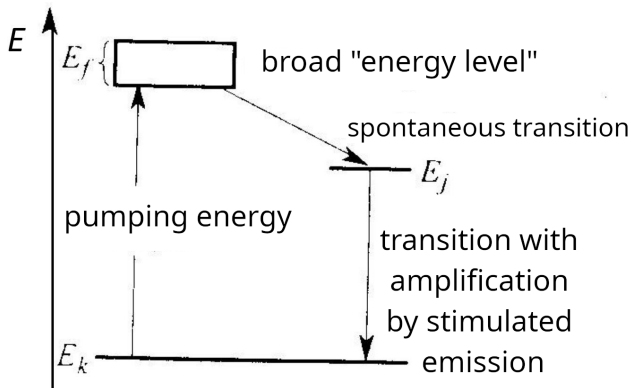
A stimulating external photon can “knock” electrons out of the higher energy state, initiating emission. This is why the process is termed *stimulated emission*. The photons produced by stimulated emission are identical in wavelength and are mutually coherent (their phases are the same).

Absorption, Emission, Stimulated Emission, LASER IV

- ▶ Microwave Amplification by Stimulated Emission of Radiation (MASER): Devices that amplify microwaves through stimulated emission and subsequently emit microwave radiation are called masers (from the corresponding English acronym).
- ▶ Light Amplification by Stimulated Emission of Radiation (LASER): Devices that amplify light through stimulated emission and emit coherent light waves are called lasers (from the English acronym).
- ▶ The first maser was built in 1950; the first laser - a ruby laser - was constructed by Theodore Maiman in 1960.
- ▶ Stimulated emission was predicted by Einstein.

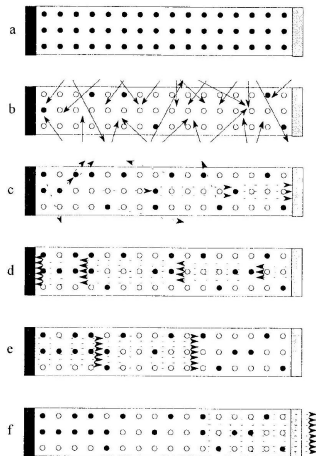
Absorption, Emission, Stimulated Emission, LASER V

- Laser Energy Scheme: A genuine two-level laser cannot operate. Pumping excites atoms; spontaneous and stimulated emissions amplify the radiation within a resonator until the light escapes through a partially transmitting mirror.



Absorption, Emission, Stimulated Emission, LASER VI

► Formation of a Light Avalanche

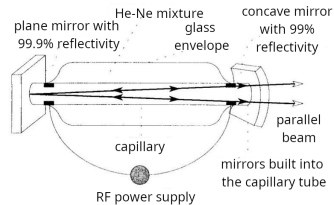


- The medium initially contains ions or atoms in the ground state.
- Pumping light (arrows) excites most atoms or ions into higher energy states (empty circles).
- Excited atoms or ions spontaneously (i.e. without external influence) emit photons along the resonator axis (dashed arrows).
- These photons multiply through stimulated emission and furthermore. . .
- . . . after repeated reflections from the front and back mirrors, the process intensifies in an avalanche-like manner.
- Once the beam becomes sufficiently strong, it exits the laser tube through the partially transmitting (99%) mirror $\Rightarrow$ laser.

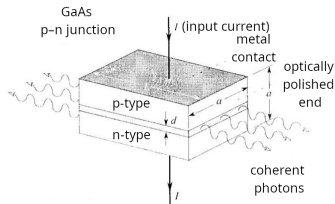
Absorption, Emission, Stimulated Emission, LASER VII

► Laser Examples

Helium–neon laser



Semiconductor lasers.



Typical values:

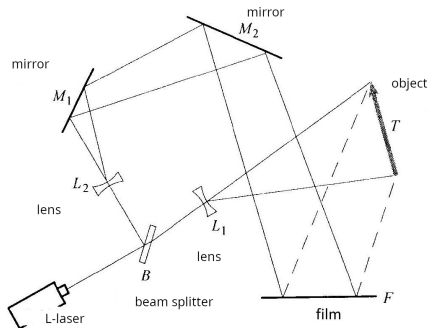
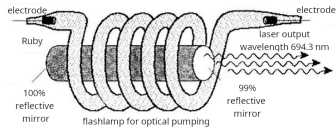
$L = 0.3 \text{ mm}$
 $a = 0.1 \text{ mm}$
 $d = 2.0 \mu\text{m}$
 $\lambda = 840 \text{ nm}$

$I \text{ (300 K-ne)} = 20 \text{ A}$
output power = 136 W
optical output power = 9 W

Absorption, Emission, Stimulated Emission, LASER VIII

► Laser Examples and Applications

- Ruby laser
- Holography: Invented by Denis Gabor. Interference between the object beam and a reference beam produces a complex interference pattern recorded on film, forming a hologram. The hologram itself *is* the interference pattern.



The End

Thank you for your attention!